Hurricane Application Scenarios

Hurricane Scarlett has hit! She made landfall near Charleston, SC, and is coming in strong. (Don't worry—it's pretend!)

The Carter family decided to wait out the storm in their downtown home. Things aren't looking so good. The power has gone out, leaving the house hot and humid, and their daughter Sophie's asthma symptoms have worsened. Her parents need help getting her to a shelter with medical staff.

Floodwater is rising around their home, and they cannot safely use their car. They need transportation for two adults, Sophie, and their dog, Buddy. They refuse to leave Buddy behind.

Create the relief request

Have Sophie's father create an account and submit a relief request. The request must include the family's location, contact information, number of people needing transportation, Sophie's need for medical assistance, and the fact that they are bringing a dog.

Choose fictional names and ages for both parents, an age for Sophie, and a Charleston home address to use throughout the scenario.

Match Rosa with the request

Meanwhile, Rosa Delgado is using her Ford F-350 truck to help neighbors reach shelters. Her volunteer profile lists transportation as her specialty.

When Rosa logs into the system, she sees seven open relief requests. The Carter family's request appears at the top because it is nearby and matches her specialty. The app clearly identifies it as a recommended request and explains why it is recommended.

Rosa opens the request and sees that two adults, one child, and one dog need transportation, and that Sophie needs medical assistance.

Find a shelter and assign the request

Rosa uses the app to search the ten shelters in the system. She needs the shelter closest to the Carter family's home by driving distance that:

- Accepts dogs.
- Has medical staff available.
- Has space for all three family members and Buddy.

After selecting a suitable shelter, Rosa assigns the relief request to herself. Its status changes to "Assigned," and other volunteers can see that Rosa is handling it.

Export the trip information

Rosa uses the app to export a clearly formatted `.txt` file that she can print. It includes:

- The family members' names and ages, plus Buddy's name and identification as a dog.
- The family's contact information and pickup address.
- A note about Sophie's need for medical assistance.
- Directions from Rosa's location to the Carter home.
- Directions from the Carter home to the selected shelter.
- The shelter's name and address.

For this fictional scenario, assume a safe, passable route is available.

Demonstrate role permissions

After transporting the family to the shelter, Rosa attempts to mark the relief request as "Complete." The system prevents the change and explains that her volunteer role does not have permission to complete relief requests. The request remains assigned to her.

Check the family into the shelter

Felix, the shelter manager, logs into the system. He can view his shelter's name, location, total capacity, current occupancy, available spaces, and list of occupants.

Felix checks the Carter family and Buddy into the shelter. The system:

- Adds the three family members to the occupant list.
- Increases human occupancy by three.
- Records Buddy as an accompanying pet.

Felix then marks the relief request as "Complete." Unlike Rosa, his shelter manager role has permission to perform this action.

Show that the changes are saved

Show the relevant JSON files before and after the scenario. Make it clear where the system saves the new account, relief request, assignment, shelter check-in, pet information, and completed request status. Demonstrate that these changes remain after restarting the app.